CS612 Lecture 13 — In-Class Exercises

NP-Completeness, Reductions & Exact Exponential Algorithms · Sat 19 Dec 2026 · ungraded

Opening

"No two conflicting nurses share a shift slot; minimise the number of slots." Which classic problem is this? How would you convince a manager that no efficient exact algorithm is known?

A — Certificates and directions

1. Give a certificate and a verifier (with running time) for: (a) G has a clique of size $\geq k$; (b) the integer N is composite; (c) a knapsack instance has a solution of value $\geq V$.
2. True or false, with a one-line reason: (a) $A \leq_p B$ and A NP-hard ⇒ B NP-hard; (b) $A \leq_p B$ and $B \in$ P ⇒ $A \in$ P; (c) $A \leq_p B$ and B NP-hard ⇒ A NP-hard; (d) every NP-hard problem is in NP; (e) an $O(2^{\sqrt{n}})$ algorithm for an NP-complete problem would imply P = NP.

B — Reductions

1. **3-COLOR ≤ 4-COLOR.** Construction and both directions.
2. **HAM-CYCLE ≤ TSP** (decision). Construction. What does it imply about approximating TSP without the triangle inequality?
3. **Find the flaw:** "LONGEST-PATH is NP-hard because I reduced it to 3-SAT."

C — Exact methods

1. Why is "stop when the partial sum exceeds T" invalid for subset sum with negative numbers? Give an instance.
2. Prove that the fractional-knapsack bound is a valid branch-and-bound bound.
3. Derive the recurrence and the base of the exponent for vertex-cover branching on a vertex v of degree $d \geq 3$ ("v in the cover, or all of $N(v)$").
4. Estimate the running time of Held–Karp for $n = 20$ and $n = 40$.